EE5311: Digital IC Design
Tutorial 3 Report

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Roll Number: EE22B032

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Question 1

Part A

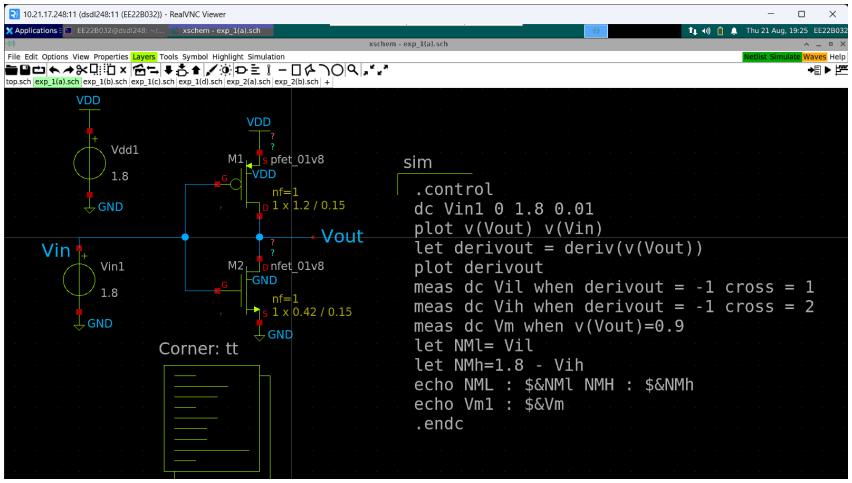


Figure 1: Schematic

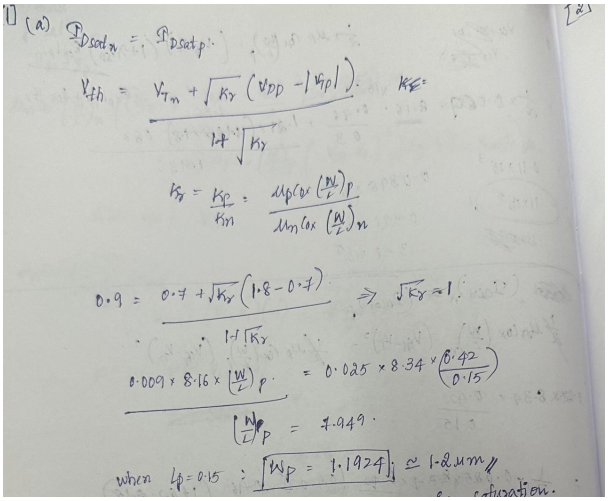


Figure 2: Width obtained

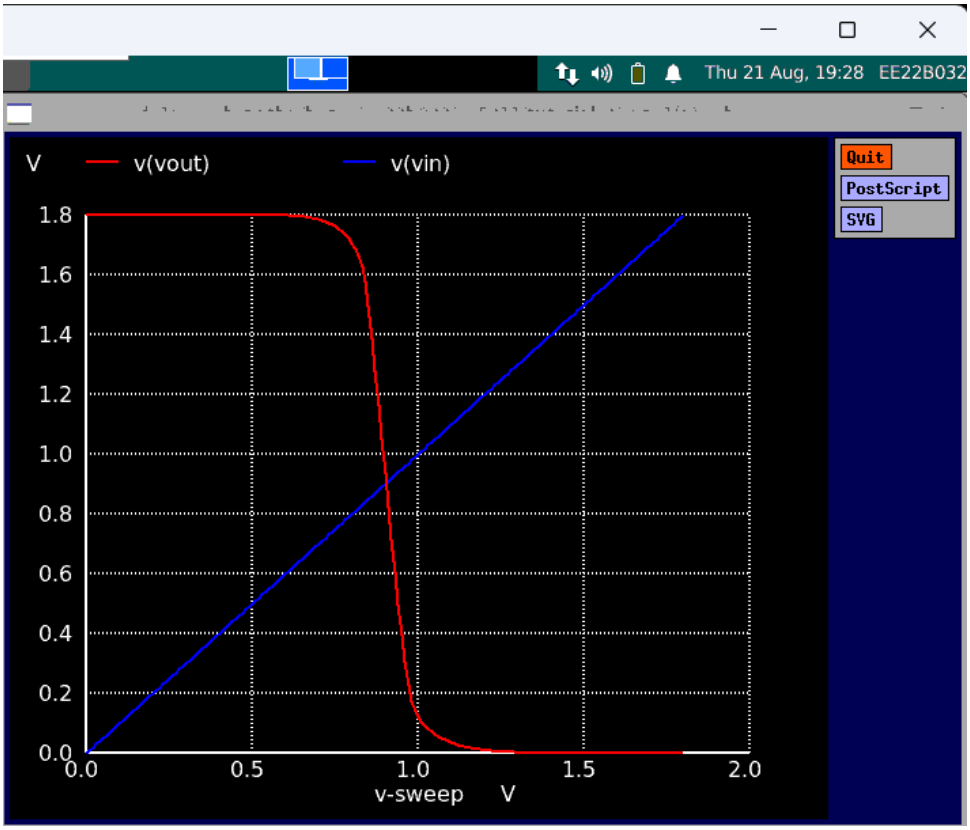


Figure 3: DC Transfer Characteristics

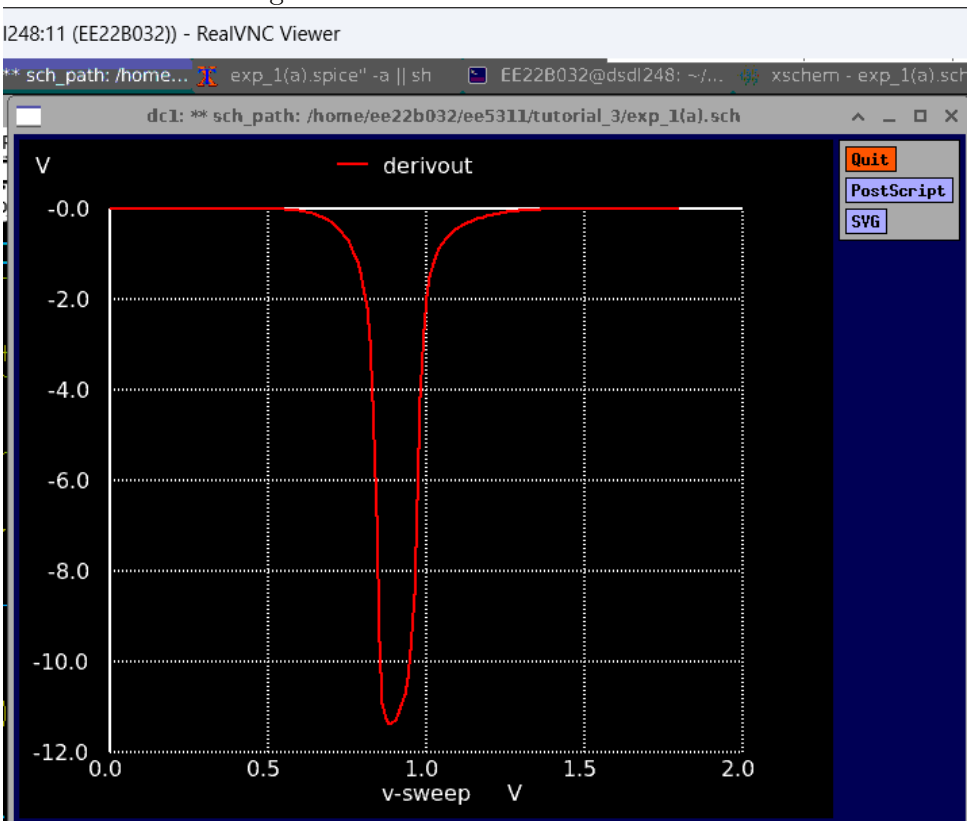


Figure 4: Current Derivative



Figure 5: Noise Margins

Part B

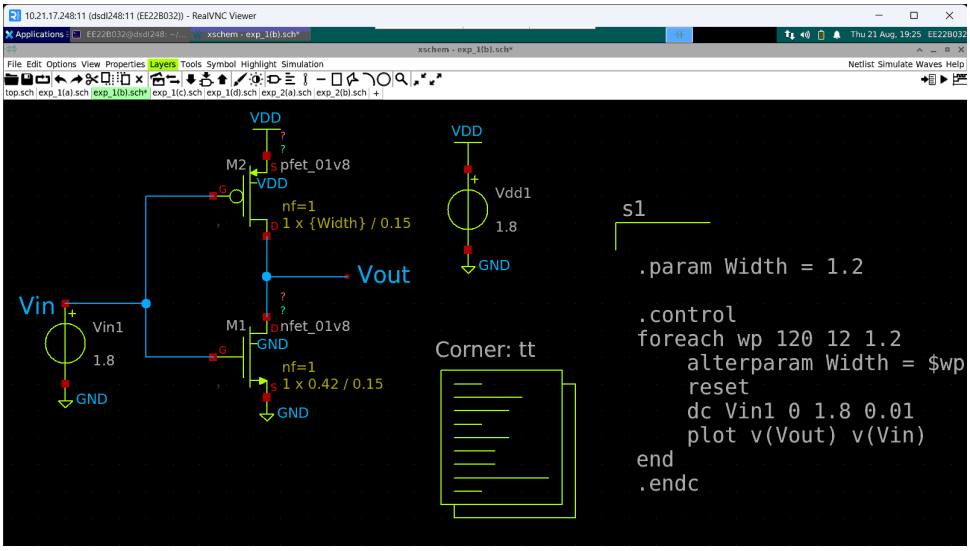


Figure 6: Schematic

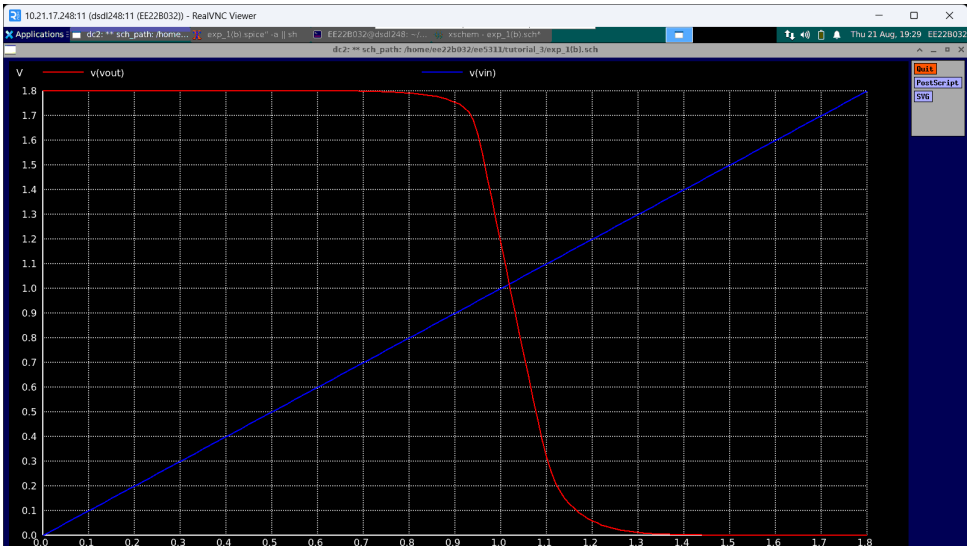


Figure 7: Original Width

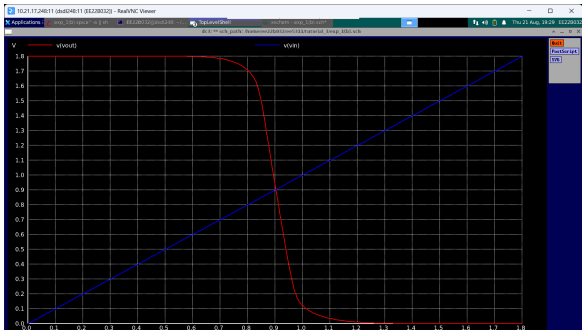


Figure 8: Increased by 10

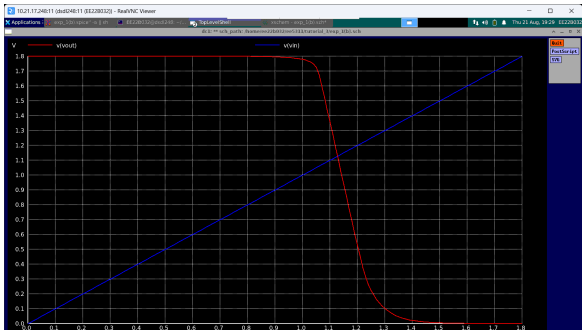


Figure 9: Decreased by 10

By increasing $(W/L)_p$ by a factor of 10:

- The PMOS transistor becomes stronger (higher drive current for same V_{GS}).
- Transition region shifts towards lower V_{in}.
- Slope of DC characteristics becomes steeper.

By decreasing $(W/L)_p$ by a factor of 10:

- The PMOS transistor becomes weaker (lower drive current for same V_{GS}).
- Transition region shifts towards higher V_{in}.
- Slope of DC characteristics becomes less steep.

Part C

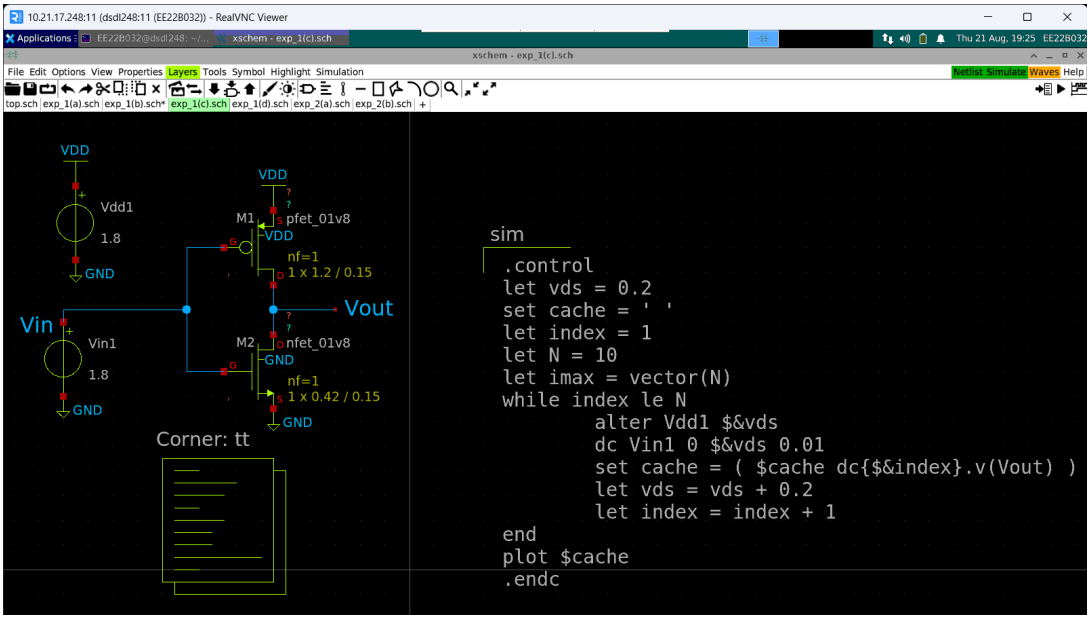


Figure 10: Schematic

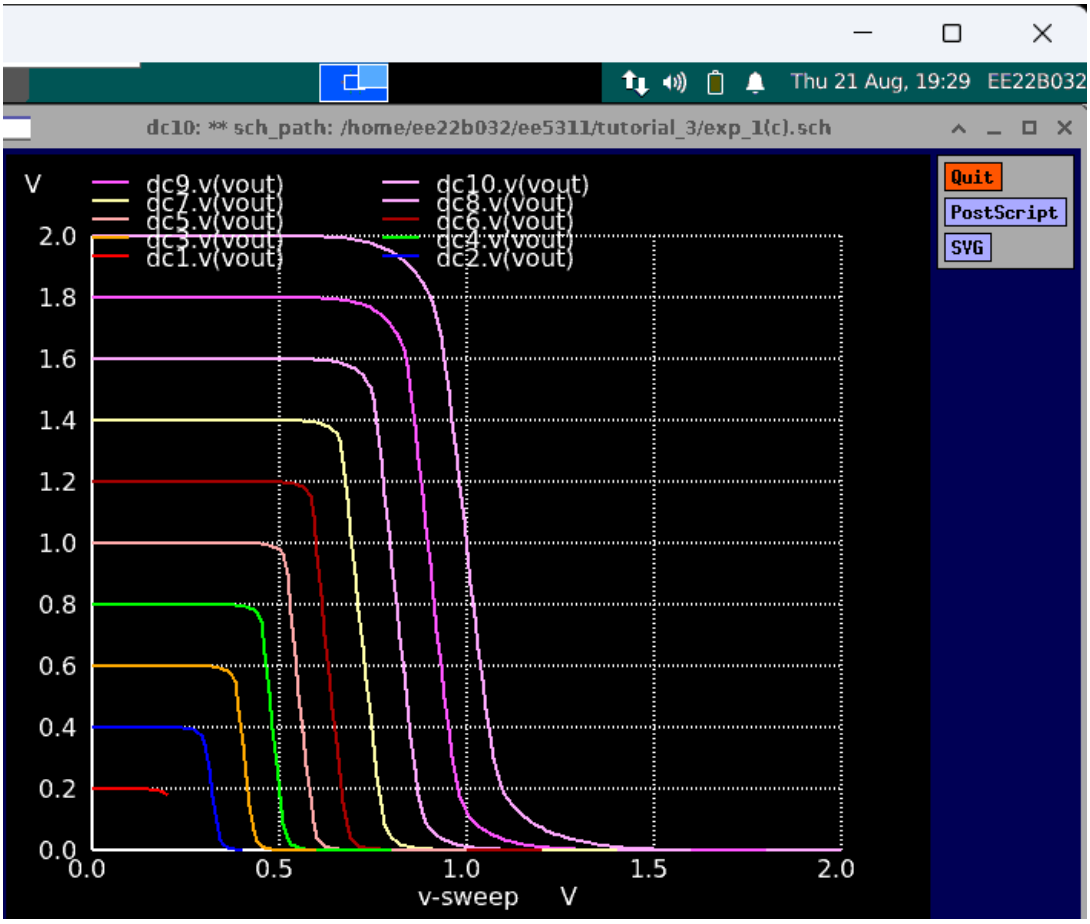


Figure 11: DC transfer Characteristics

Part D

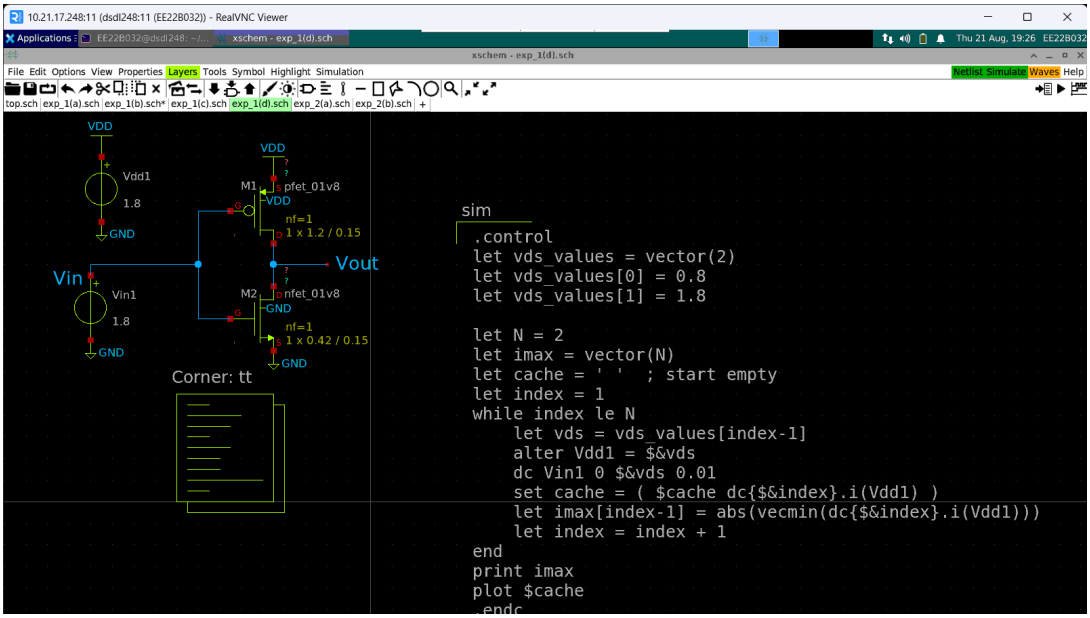


Figure 12: Schematic

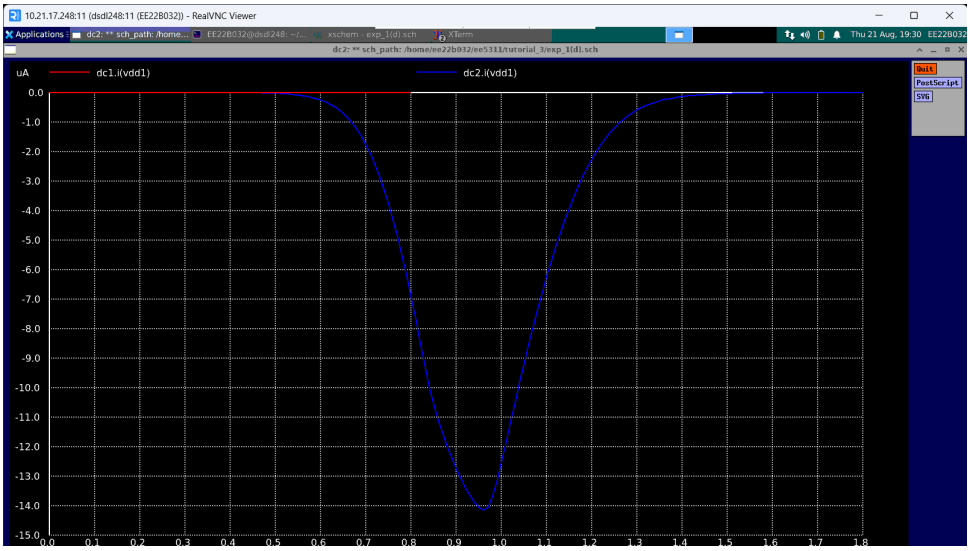


Figure 13: IDS

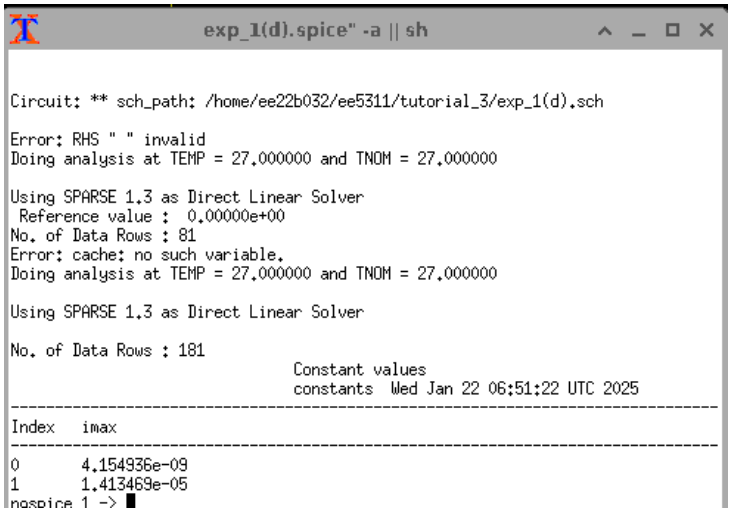


Figure 14: Ipeak simulated

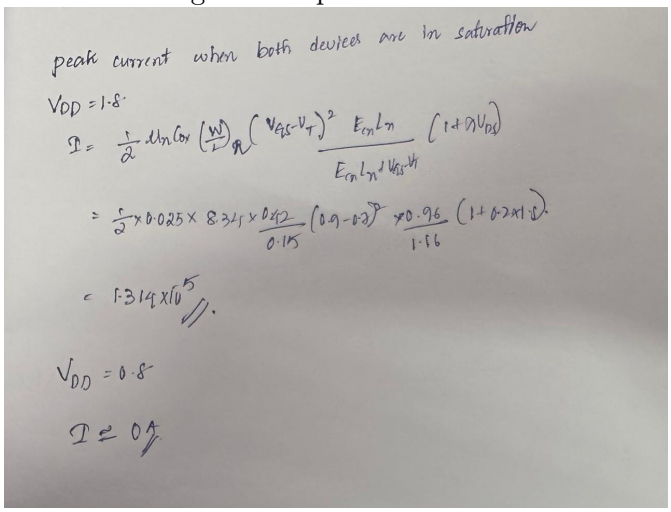


Figure 15: Ipeak Analytical

Question 2

Part A

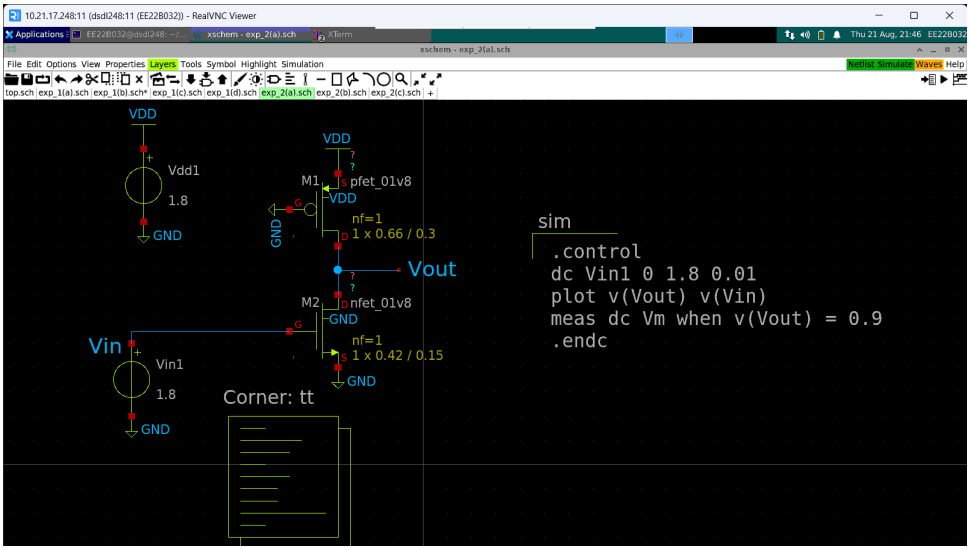


Figure 16: Schematic

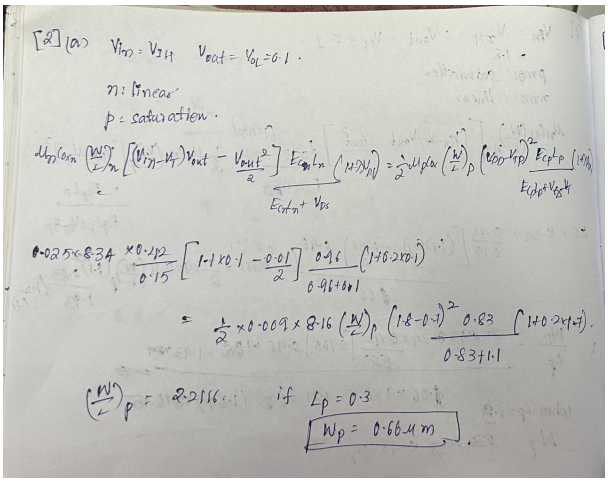


Figure 17: Width

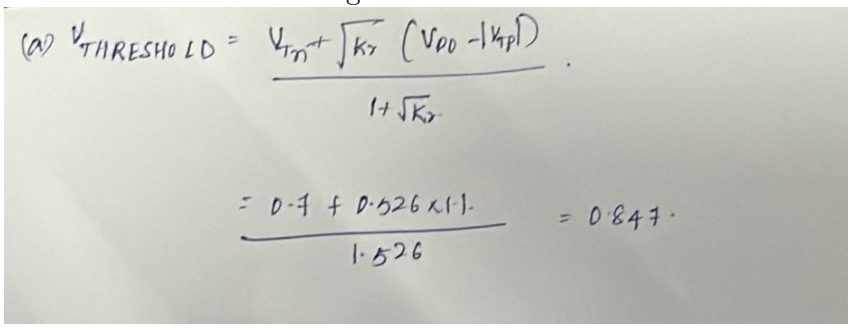


Figure 18: Threshold voltage

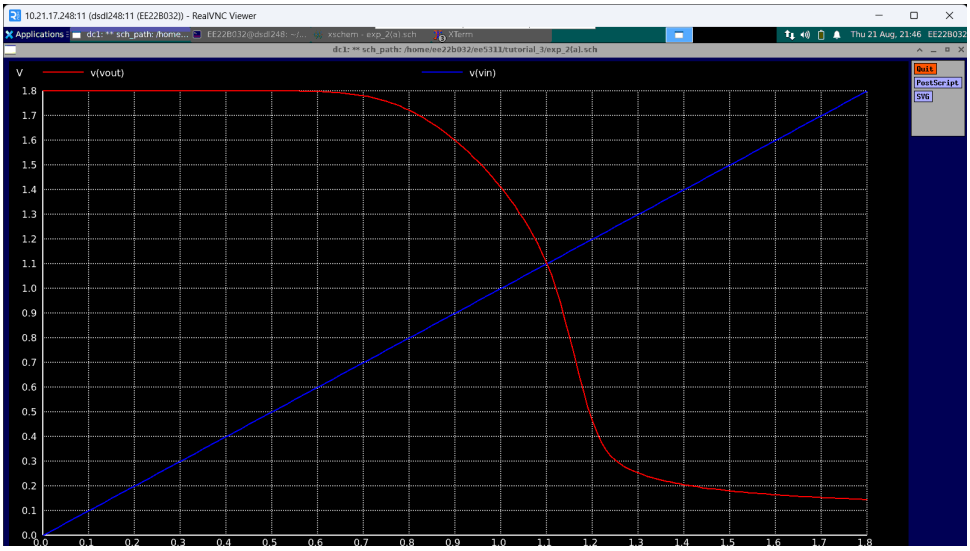
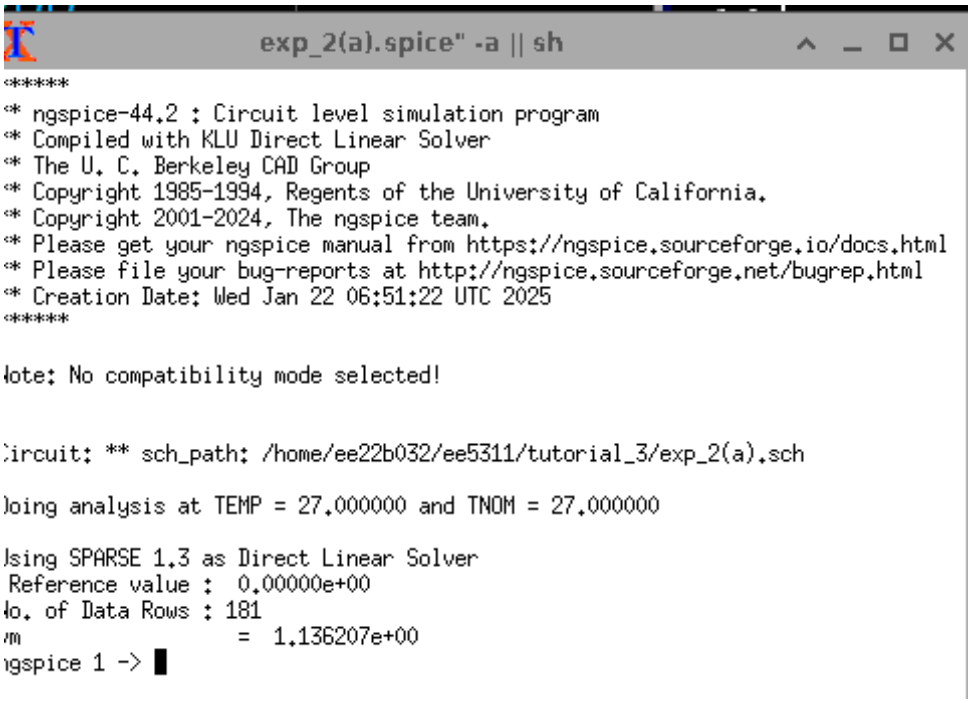


Figure 19: DC Characteristics



Part B

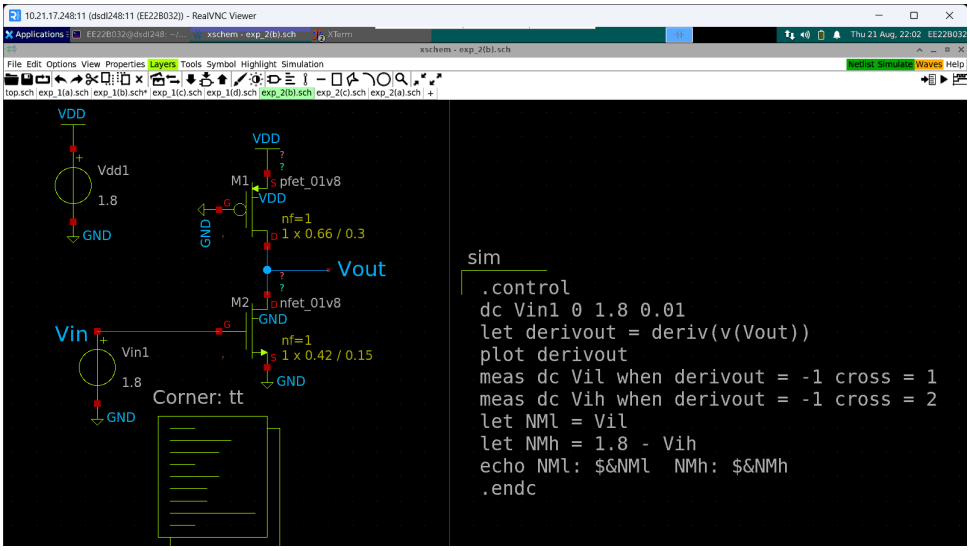


Figure 20: Schematic

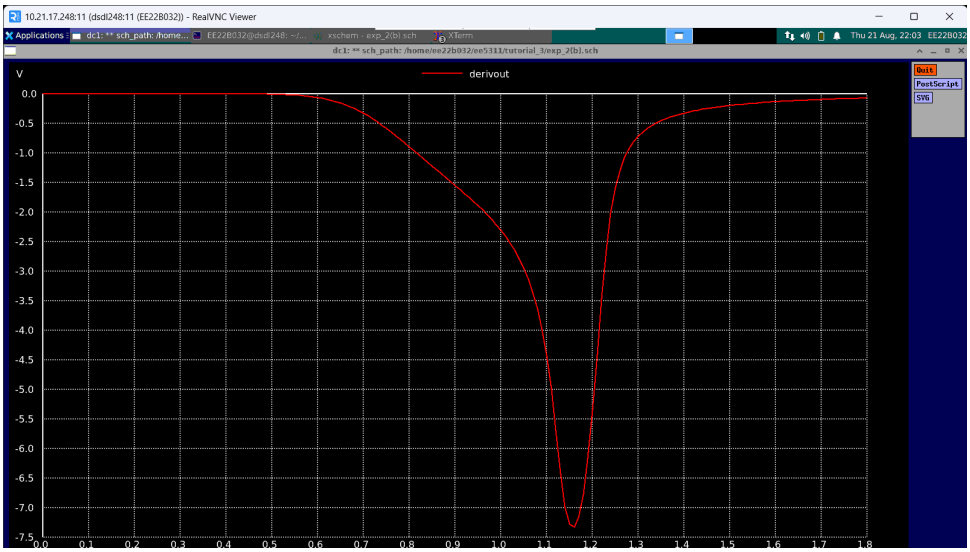


Figure 21: Current Derivative

Part C

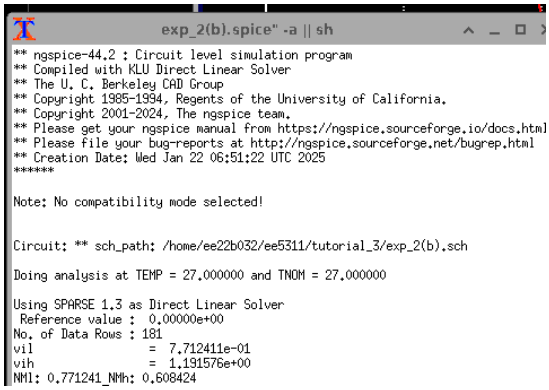


Figure 22: Noise Margin Simulated

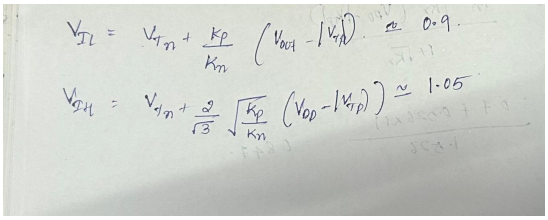


Figure 23: Noise Margin Analytical

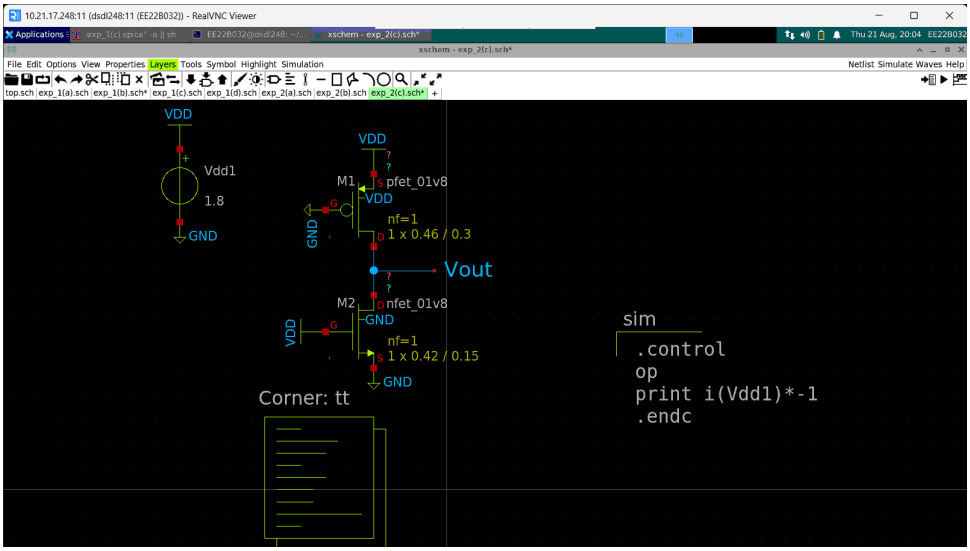


Figure 24: Schematic

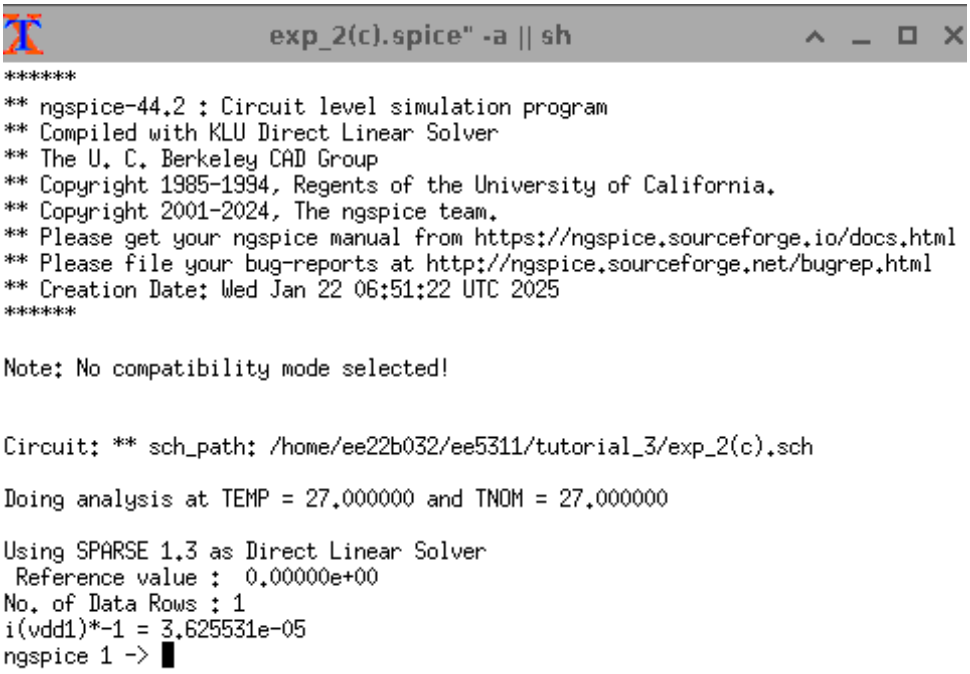


Figure 25: Simulated current

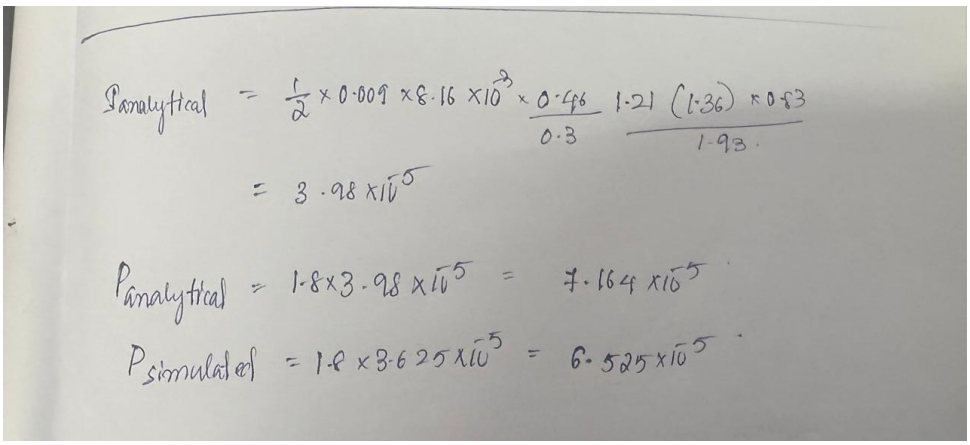


Figure 26: Analytical values